Data Dictionary & Schema Specification

Document Code: DD-DOC-001

Project: Duplicate Bin Ecosystem

Databases: Cloud SaaS Database (SQLAlchemy / PostgreSQL / SQLite) & Client Persistent Cache (SQLite WAL)

1. Cloud SaaS Database (`backend/app/models.py`)

1.1 Table: `users`

Represents customer account credentials, subscription tiers, and credit ledgers.

Field Name	Data Type	Nullable	Primary Key	Default	Business Description & Validation Rules	Example
`id`	`VARCHAR(36)`	No	Yes	`uuid.uuid4()`	System-generated unique identifier for the user account.	`"c8a9f3b2-7a41-48e2-b1c5-23f4a908d1e3"`
`email`	`VARCHAR(255)`	No	No	None	Unique login email address. Must be a valid email format.	`"elena.photos@gmail.com"`
`name`	`VARCHAR(100)`	No	No	None	Full display name of the customer or	`"Elena Rostova"`

					customer or	

Field Name	Data Type	Nullable	Primary Key	Default	Business Description & Validation Rules	Example
					account holder.	
`password_hash`	`VARCHAR(255)`	No	No	None	Bcrypt-hashed password digest. Plaintext is never persisted.	`"$2b$12$e8Y4vL..."`
`credit_bytes`	`BIGINT`	Yes	No	`10737418240`	Available deduplication credit balance in bytes. Default 10 GB. Debited upon scan completion.	`10737418240` (10 GB)
`plan`	`VARCHAR(50)`	Yes	No	`"SaaS Premium"`	Active subscription tier (e.g., "Free", "SaaS Premium", "Team Starter", "Business Premium").	`"Team Starter"`

Field Name	Data Type	Nullable	Primary Key	Default	Business Description & Validation Rules	Example
`created_at`	`DATETIME`	Yes	No	`utcnow()`	UTC timestamp when account was registered.	`"2026-10-03 14:32:00"`

1.2 Table: `devices`

Represents paired client machines running the Duplicate Bin background daemon.

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
`id`	`VARCHAR(36)`	No	Yes	No	Unique hardware identifier or client-generated machine UUID.	`"agent-node-04-macbook"`
`user_id`	`VARCHAR(36)`	No	No	`users.id` (CASCADE)	Identifies the owning user account. Deleted if user account	`"c8a9f3b2-7a41-..."`

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
					user account is deleted.	
`name`	`VARCHAR(100)`	No	No	No	Hostname or user-assigned label for the machine.	`"Elena-Studio-MacBook"`
`os`	`VARCHAR(50)`	No	No	No	Operating system platform (`Windows`, `Linux`, `Darwin`).	`"Windows 11 Pro"`
`pairing_token`	`VARCHAR(64)`	No	No	No	64-character hexadecimal shared secret used to authenticate WebSockets.	`"9a4f2c018b7e..."`
`is_active`	`BOOLEAN`	Yes	No	No	Online presence flag. Dynamically synchronized with active WebSockets.	`True`
`last_seen`	`DATETIME`	Yes	No	`utcnow()`	Timestamp of	`"2026-10-03`

					Timestamp of	`"2026-10-03`

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
					last received telemetry heartbeat or WebSocket ping.	`18:15:22"`
`created_at`	`DATETIME`	Yes	No	`utcnow()`	Initial registration timestamp of the device.	`"2026-10-01 09:12:00"`

1.3 Table: `scans`

Records completed or in-progress scan audits executed by paired devices.

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
`id`	`VARCHAR(36)`	No	Yes	No	Unique scan audit identifier.	`"scan-8291f0-4821"`
`device_id`	`VARCHAR(36)`	No	No	`devices.id` (CASCADE)	Identifies the physical device executing the scan.	`"agent-node-04-macbook"`

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
`status`	`VARCHAR(20)`	No	No	No	Lifecycle status: `running`, `completed`, `failed`, `cancelled`.	`"completed"`
`path_scanned`	`VARCHAR(500)`	No	No	No	Absolute root path evaluated during this scan audit.	`"D:\Photos\2026_Weddings"`
`files_count`	`INTEGER`	Yes	No	`0`	Total number of files scanned in target directory.	`14250`
`duplicates_count`	`INTEGER`	Yes	No	`0`	Total redundant duplicate files identified.	`3120`
`bytes_recoverable`	`BIGINT`	Yes	No	`0`	Total potential storage savings in bytes. Used	`48318382080` (~45 GB)

					bytes. Used	

Field Name	Data Type	Nullable	Primary Key	Foreign Key	Business Description & Validation Rules	Example
					to debit `credit_bytes`.	
`duration_seconds`	`FLOAT`	Yes	No	`0.0`	Total runtime of the scan in seconds.	`24.8`
`created_at`	`DATETIME`	Yes	No	`utcnow()`	Timestamp when scan was started or commanded.	`"2026-10-03 14:00:00"`
`completed_at`	`DATETIME`	Yes	No	None	Timestamp when scan finished and telemetry was ingested.	`"2026-10-03 14:00:25"`

2. Desktop Local Hash Cache Database (`hash_cache.db`)

Embedded SQLite database stored in `~/.cache/duplicate-bin/hash_cache.db`.

Column Name	SQLite Type	Constraints	Business Description
`file_path`	`TEXT`	`PRIMARY KEY`	Absolute canonical file path on local filesystem.
`file_size`	`INTEGER`	`NOT NULL`	Exact size in bytes. Checked against disk

			Exact size in bytes. Checked against disk

Column Name	SQLite Type	Constraints	Business Description
			`st_size` on cache lookup.
`mtime`	`REAL`	`NOT NULL`	POSIX modification timestamp. If disk `mtime > cache.mtime`, cache is invalidated.
`hash_value`	`TEXT`	`NOT NULL`	Cryptographic hex digest (SHA-256 / MD5).
`algorithm`	`TEXT`	`NOT NULL`	Algorithm name (`sha256`, `md5`, `blake2b`).
`cached_at`	`TIMESTAMP`	`DEFAULT CURRENT_TIMESTAMP`	Timestamp when hash was recorded.

3. Desktop Quarantine Manifest Format (`manifest.json`)

Stored locally inside each quarantine batch folder `~/.duplicate_bin/archive/<batch_timestamp>/manifest.json`.

```
{
  "batch_id": "20261003_140522",
  "created_at": "2026-10-03T14:05:22.182Z",
  "total_files": 3,
  "total_bytes_archived": 104857600,
  "items": [
    {
      "original_path": "C:\\Users\\User\\Pictures\\Project\\shot_copy.raw",
      "archive_relative_path": "Pictures\\Project\\shot_copy.raw",
      "file_size": 52428800,
      "sha256": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"
    }
  ]
}
```